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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

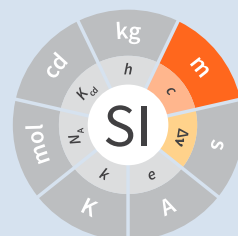
Part 2

Mercury 199 Atom ()

Mercury 199 Atom ()

CCL-CCTF Frequency Standards Working Group

August 2018
août 2018



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Mercury 199 Atom ()**

CCL-CCTF Frequency Standards Working Group

9th edition 2019

1. Recommended value [1] of the frequency in the CIPM List of Frequencies [2]

$$f(^{199}\text{Hg}) = 1\,128\,575\,290\,808\,154.4\text{ Hz}$$

equivalent to

$$\lambda(^{199}\text{Hg}) = 265\,637\,977.759\,838\,7\text{ fm},$$

with a relative standard uncertainty of 5×10^{-16} .

This radiation was endorsed by the CCTF [1] as a secondary representation of the definition of the second.

2. Source data

The adopted value of $f(^{199}\text{Hg}) = 1\,128\,575\,290\,808\,154.4\text{ Hz}$ with a relative standard uncertainty of 5×10^{-16} has been calculated by the methods presented in [3], [4], [5], [6], with the data from the table of evaluated frequencies (2017) attached below. The estimated relative standard uncertainty of 5×10^{-16} takes into account the relative uncertainty of the ^{87}Sr value of 4×10^{-16} governing the evaluation procedure and quantitatively unknown contributions of the correlations between the data of [8].

Table. **Table of evaluated frequencies (2017)**

Frequency / Hz	Uncertainty / Hz	Reference
^{199}Hg		
1 128 575 290 808 155.1	6.4	[7]
1 128 575 290 808 154.62	0.41	[8]
^{171}Yb		
518 295 836 590 864.0	28.0	[38]
518 295 836 590 863.1	2.0	[39]
518 295 836 590 865.2	0.7	[40]
518 295 836 590 863.5	8.1	[41]
518 295 836 590 863.59	0.31	[42]
518 295 836 590 863.38	0.57	[43]
^{88}Sr		
429 228 066 418 009.0	32.0	[9]
429 228 066 418 008.3	2.1	[10], Sr1
429 228 066 418 007.3	2.9	[10], Sr2
^{87}Sr		
429 228 004 229 874.0	1.1	[11]
429 228 004 229 873.65	0.37	[12]
429 228 004 229 873.6	1.1	[13]
429 228 004 229 874.1	2.4	[14]
429 228 004 229 872.9	0.5	[15]
429 228 004 229 873.9	1.4	[16]
429 228 004 229 872.0	1.6	[17]
429 228 004 229 873.56	0.49	[18]
429 228 004 229 873.7	1.4	[19]
429 228 004 229 873.13	0.17	[20]
429 228 004 229 873.10	0.13	[21]
429 228 004 229 872.92	0.12	[22]
429 228 004 229 872.97	0.16	[23], 2014
429 228 004 229 873.04	0.11	[23], June 2015

Frequency / Hz	Uncertainty / Hz	Reference
429 228 004 229 872.97	0.40	[24]
429 228 004 229 872.99	0.18	[25]
$^{40}\text{Ca}^+$		
411 042 129 776 393.2	3.0 ^(a)	[26]
411 042 129 776 398.4	1.2	[27]
411 042 129 776 400.5	1.2	[28], 2012
411 042 129 776 401.7	1.1	[28], 2014/2015
^{87}Rb		
6 834 682 610.904 312 9	2.4×10^{-6}	[29]
6 834 682 610.904 307 0	3.1×10^{-6}	[30]
6 834 682 610.904 312 5	2.1×10^{-6}	[31]
Frequency ratios	Fractional uncertainty	Reference
$^{199}\text{Hg}/^{87}\text{Sr}$		
2.629 314 209 898 909 60	8.4×10^{-17}	[32]
2.629 314 209 898 909 15	1.8×10^{-16}	[8]
$^{199}\text{Hg}/^{87}\text{Rb}$		
165 124.754 879 997 258	3.8×10^{-16}	[8]
$^{171}\text{Yb}/^{87}\text{Sr}$		
1.207 507 039 343 341 2	1.4×10^{-15}	[33], [34]
1.207 507 039 343 337 76	2.4×10^{-16}	[35]
1.207 507 039 343 337 749	4.6×10^{-17}	[36]
$^{88}\text{Sr}/^{87}\text{Sr}$		
1.000 000 144 883 682 777	$7 \times 10^{-17} \text{ }^{(b)}$	[37]
$^{87}\text{Sr}/^{40}\text{Ca}^+$		
1.044 243 334 529 641 6	2.3×10^{-15}	[27]
$^{87}\text{Sr}/^{87}\text{Rb}$		
62 801.453 800 512 435	3.3×10^{-16}	[22]
(a) From the least square procedure it turned out that the value and the uncertainty given in ref. [26] is not compatible with the remaining data. Thus the uncertainty given in the original publication [26] was increased to 3 Hz to make it statistically more consistent. (b) The fractional uncertainty of 2.3×10^{-17} in ref. [37] has been increased by a factor of three since this uncertainty is an order of magnitude smaller than the other input data for this transition.		

^{199}Hg neutral atom, $6s^2\ ^1S_0 - 6s6p\ ^3P_0$ unperturbed optical transition

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